

BREGENZ, Austria

WMO: 111010

Lat: 47.4992N

Lon: 9.7461E

Elev: 424

StdP: 96.34

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.1	-10.9	1.6	-7.5	-8.7	1.9	-5.1	6.9	4.6	6.2	4.3	1.5	190	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	29.1	20.2	27.3	19.7	25.8	19.0	21.3	27.2	20.4	25.9	19.6	24.6	2.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.7	24.1	18.5	14.0	23.2	17.7	13.4	22.3	63.6	27.5	60.5	26.0	57.7	24.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.3	5.4	4.6	-10.2	32.4	2.5	2.4	-12.0	34.1	-13.5	35.5	-14.9	36.8	-16.6	38.5		
			-10.7	23.1	2.4	0.9	-12.4	23.7	-13.8	24.3	-15.2	24.8	-17.0	25.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	1.3	2.3	6.2	10.2	14.5	18.0	19.6	19.2	14.9	10.9	5.6	2.2
	DBStd	7.54	4.11	4.17	3.85	4.06	3.70	3.86	3.36	3.17	3.23	3.44	3.83	3.84
	HDD10.0	1084	271	217	128	48	6	1	0	0	2	31	139	242
	HDD18.3	3062	529	449	378	246	129	53	26	27	111	232	382	500
	CDD10.0	1246	1	1	9	52	144	241	297	287	149	58	7	1
	CDD18.3	181	0	0	0	1	9	43	64	56	8	1	0	0
	CDH23.3	1047	0	0	0	5	52	270	398	298	23	1	0	0
	CDH26.7	220	0	0	0	1	6	59	91	63	1	0	0	0
Wind	WSAvg	1.9	1.9	1.8	1.9	1.9	1.9	2.0	1.9	1.8	1.7	1.7	1.9	
Precipitation	PrecAvg	1400	69	77	88	101	138	171	168	170	124	96	104	91
	PrecMax	1799	150	210	177	188	321	314	311	291	200	208	238	192
	PrecMin	1067	7	21	30	32	19	65	65	49	57	9	7	10
	PrecStd	155	40	41	37	42	67	61	65	60	39	43	50	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	14.0	18.0	23.6	27.3	31.1	31.5	31.0	25.7	21.3	17.8	14.8
		MCWB	8.3	8.6	10.9	14.0	17.8	20.6	21.1	20.9	18.7	15.2	11.5	8.3
	2%	DB	10.1	11.3	15.5	21.1	24.7	28.4	29.1	28.3	23.5	18.8	14.6	10.3
		MCWB	7.0	7.1	9.9	13.2	16.8	19.8	20.3	20.7	18.0	14.5	10.4	7.8
	5%	DB	8.4	9.5	13.6	18.8	22.8	26.4	27.2	26.5	21.6	17.1	12.2	8.8
		MCWB	6.0	6.3	9.1	12.4	15.7	19.1	19.6	19.9	17.1	13.8	9.8	6.9
	10%	DB	6.7	7.8	11.7	16.8	20.7	24.6	25.6	24.7	19.9	15.5	10.6	7.4
		MCWB	4.8	5.3	8.2	11.2	14.8	18.2	19.0	19.3	16.3	13.0	8.8	5.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	9.1	11.8	15.2	19.0	22.0	22.4	22.5	19.7	16.3	12.5	9.3
		MCDB	11.2	12.6	16.4	21.4	25.2	29.2	28.9	28.5	23.9	19.4	16.0	12.2
	2%	WB	7.4	7.6	10.6	13.7	17.4	20.6	21.2	21.3	18.6	15.3	11.0	8.1
		MCDB	9.5	10.6	14.3	19.5	23.1	26.7	27.3	26.9	22.2	17.8	13.3	10.1
	5%	WB	6.1	6.6	9.6	12.7	16.3	19.7	20.2	20.4	17.6	14.2	10.0	7.1
		MCDB	8.0	9.1	12.8	17.9	21.5	25.2	25.8	25.5	21.0	16.3	12.0	8.7
	10%	WB	5.0	5.5	8.5	11.7	15.3	18.7	19.4	19.5	16.6	13.4	9.0	5.9
		MCDB	6.5	7.6	11.3	16.0	19.7	23.7	24.5	24.0	19.4	15.2	10.5	7.2
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.5	7.1	8.2	8.2	8.3	8.4	8.2	7.3	5.9	4.8	4.3
		MCDBR	6.2	8.2	10.2	11.7	11.4	11.3	11.1	11.0	9.7	8.1	7.6	6.3
		MCWBR	4.2	5.1	5.7	5.8	5.1	5.0	4.8	5.0	4.5	4.4	4.1	
	5% WB	MCDBR	5.8	7.4	9.1	10.7	10.0	10.2	10.3	10.1	8.7	7.0	6.5	5.9
		MCWBR	4.3	4.8	5.3	5.5	4.9	4.8	4.9	5.1	4.8	4.3	4.2	4.0
Clear-Sky Solar Irradiance	taub	0.288	0.315	0.355	0.398	0.413	0.419	0.420	0.412	0.381	0.348	0.316	0.285	
	taud	2.406	2.332	2.304	2.240	2.253	2.277	2.271	2.310	2.374	2.451	2.478	2.429	
	Ebn at Noon	798	845	860	853	852	847	840	831	824	800	760	757	
	Edh at Noon	68	90	109	128	132	129	129	119	102	81	64	59	
All-Sky Solar Radiation	RadAvg	1.18	2.04	3.21	4.31	4.95	5.64	5.47	4.80	3.58	2.30	1.28	0.97	
	RadStd	0.13	0.27	0.41	0.60	0.52	0.54	0.51	0.51	0.39	0.25	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.40	N/A	N/A	+1.16	N/A	N/A	-172	+117	+52	
(47)	Regional (107 neighbors)	N/A	N/A	N/A	+0.44	+0.39	N/A	N/A	+53	+13	

Nomenclature: See separate page